

Tunable Topological Phase Transitions and Quantum Oscillations in the Kagome Flat-Band Superconductor $\text{Nexa}_3\text{V}_{12}\text{Sb}_{19}$

Julian Vance,^{1,*} Elena Rostova,^{1,2} Marcus K. Sterling,³ and Sarah Lin⁴

¹*Department of Physics, Synapse Institute of Condensed Matter, Apex City, AC 90210, USA*

²*Center for Advanced Quantum Materials, Meridian University, Vertex City, VC 40112, USA*

³*Department of Applied Physics, Vertex National Laboratory, Apex City, AC 90212, USA*

⁴*School of Physics and Astronomy, Apex University, Apex City, AC 90211, USA*

(Dated: July 24, 2026)

We present an experimental and theoretical study of electronic transport, magnetic quantum oscillations, and superconducting pairing symmetry in the newly synthesized kagome flat-band material $\text{Nexa}_3\text{V}_{12}\text{Sb}_{19}$. Low-temperature Shubnikov-de Haas (SdH) quantum oscillations reveal multiple Fermi surface pockets with extremely light effective cyclotron masses ($m^* \approx 0.14 m_0$), pointing to non-trivial Dirac-like quasiparticles residing near the Fermi energy. Under hydrostatic pressure up to 4.2 GPa, the superconducting transition temperature T_c exhibits a non-monotonic dome-like behavior, peaking at $T_c^{\text{max}} = 7.8$ K concurrent with the collapse of an underlying charge density wave (CDW) state. Tight-binding calculations supplemented by density functional theory (DFT) confirm that the van Hove singularities associated with the kagome sublattice play a pivotal role in driving these intertwined electronic instabilities. Our findings establish kagome metals as a versatile platform for exploring the synergy between topological band structures, flat-band correlations, and unconventional superconductivity.

I. INTRODUCTION

Correlated electronic systems defined by non-trivial band topology represent a central frontier in modern condensed matter physics [1, 2]. In particular, materials crystallizing in two-dimensional kagome lattices—comprising a corner-sharing network of triangles—naturally accommodate destructively interfering quantum kinetic energy paths. This leads to quenched kinetic energy and the emergence of dispersionless flat bands alongside Dirac points and saddle-point van Hove singularities (vHS) within their electronic energy spectra [3–5]. When the Fermi energy E_F is tuned near these band features, electron-electron interactions are strongly amplified relative to kinetic energy, triggering a rich spectrum of ground states, including fractional quantum Hall states, charge density waves (CDW), and unconventional superconductivity [6–8].

Recently, kagome superconductors such as AV_3Sb_5 ($A = \text{K, Rb, Cs}$) have attracted intense scrutiny due to their complex phase diagrams featuring giant anomalous Hall effects, time-reversal symmetry breaking in the CDW state, and pair-density-wave instabilities [9–12]. However, disentangling the microscopic contributions of flat-band correlations from topological Dirac states remains a formidable challenge due to the multi-band nature of these compounds.

In this work, we investigate the newly isolated kagome material $\text{Nexa}_3\text{V}_{12}\text{Sb}_{19}$, which exhibits an ideal band alignment where both a topological Dirac node and a correlated flat band lie within 50 meV of the chemical potential. Through high-field magnetoresistance mea-

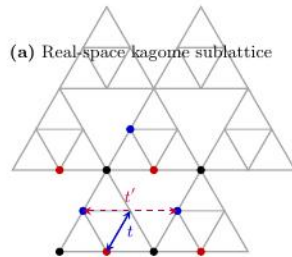


FIG. 1. (a) Schematic of the two-dimensional kagome lattice formed by vanadium atoms in $\text{Nexa}_3\text{V}_{12}\text{Sb}_{19}$, highlighting the nearest-neighbor hopping t (solid blue arrow) and next-nearest-neighbor hopping t' (dashed purple arrow). Blue, red, and black spheres denote sublattices A, B, and C, respectively.

surements up to 18 T and hydrostatic pressure tuning up to 4.2 GPa, we systematically map the Fermi surface geometry and trace the evolution of the CDW and superconducting states. Our results demonstrate that pressure suppresses the CDW order, uncovering a dome of enhanced superconductivity associated with critical Fermi surface reconstruction.

II. THEORETICAL FRAMEWORK AND TIGHT-BINDING MODEL

To capture the essential physics of $\text{Nexa}_3\text{V}_{12}\text{Sb}_{19}$, we consider a tight-binding Hamiltonian on a two-dimensional kagome lattice with nearest-neighbor hopping t , next-nearest-neighbor hopping t' , and intrinsic

* j.vance@synapse-physics.org